

Successfully Writing a BA or MA Thesis¹

At the Professorship for Computational Social Science & Big Data at TUM

0. Finding a topic

A. There are different approaches for finding a topic for a potential thesis:

1) You have an idea and you'll find an advisor who is interested in this idea.

1b) You have an external partner (other university, company, NGO, etc.) with whom you want to do your thesis.

2) You go to a potential advisor and ask if she/he has a topic for you. But even in this case it makes sense to first look at the group's projects and think about whether you can relate to one of them. Here, for instance, are our projects:

<https://www.hfp.tum.de/css/forschung/current-projects/>

All approaches are fine. The best situation is working on a topic that both you and your advisor are interested in. So, even if you have a topic in your mind, try to be flexible if potential advisors add their thoughts to your topic.

B. One more thing: Our group works with data and calculation. Consequently, all our supervised theses have to have either an empirical part including data analysis OR/AND a focus on development of computational methods.

1. Exposé

A. Before registering your thesis, you should create an exposé document. Here, you'll find details about this document, as well as about formal criteria for your thesis. These are not general rules but this just reflects the way we do things. Other groups might have different processes. The exposée should have 3-5 pages including references. The following list sounds like a lot, but: a) don't worry, nothing is lost - A good exposée can serve as a start for your thesis document. b) If you can answer these questions, the chances are very high that it will be easy for you to finish your thesis within the official time frame.

Important: The main purpose of the exposé is to make sure that your thesis will succeed. If your plans are likely to fail, it is good to figure this out before your 3/6 months clock starts to tick.

B. Here are the sections that need to go into a good exposée:

- **Problem statement:** What is the problem in the world that needs to be addressed? Why is this a problem? For whom is it a problem? Why should anyone care? What do you care about?
- **Research Question:** The problem statement is bigger, the research question (RQ) is about what specific question you want to tackle. What puzzle/problem do you want to solve? What do you want to accomplish? This is the single, core question you try to

¹ Many of these considerations also apply for writing an expose for a Ph.D. thesis - of course with more pages and more references and more time and more... :)

answer in the conclusions of your thesis. The research question is the central problem of your research around which you organize your thesis. Having a good RQ will help you to focus all your descriptions, analyses, interpretations, discussions, evaluations, and arguments.

- **Theory:** Theories are great for guiding your research. Why do you do what you do? Why would social scientists care? Where do you place your work in the scientific world? What well-defined mechanism/phenomenon do you want to understand?
- **Related Work:** You need to find 10 or more references in which researchers did something similar to what you are planning to do. Make clear, how your research is going to be similar or different to those references. Don't use too many newspaper articles, try to find scientific papers - Google Scholar will help you. Write 2-3 sentences for every reference describing what is in the article. See later in this document for how to handle references.
- **Data:** Your thesis is either based on empirical data that you are planning to collect or it is a secondary analysis of existing data or articles. What do you have? What do you want? Do you really believe that you can get all the data that you need for your thesis? How will you approach getting the data? Take into account the timeframe you would need in order to collect the data you propose. Important: Getting your data is part of the research process and can normally not be outsourced.
- **Methods:** What are you planning to do to get to your results? Which methods/algorithms are you going to use? How will you do it? Try to explain why you selected this particular method and why you think it is suitable for your RQ.
- **Outlook:** How will your results most likely look like? Why would this be great? Now, remember your problem statement and research question: How could your work answer this question or give an answer to a problem? Which target group would have which advantage after reading your thesis? Your RQ can (and sometimes also necessarily needs to) change during your research process. The outlook does not have to be the final result of your thesis, it can change as well.
- **Dissemination:** Write 3-5 lines summarizing the seven points from above in a way that a non-academic audience would understand what you are doing.

2. Registering the Thesis:

A. For registering your thesis:

1. If you are a HfP student, please use this form:
https://www.hfp.tum.de/fileadmin/w00cjd/hfp/Downloadbereich/Anmeldung_Abschlussarbeit_Registration_Final_Thesis_SOT.pdf
2. If you are a CIT student, we have to register your thesis in the online portal of the school. For this, we need your Matrikelnummer, name, language of thesis, planned title, and planned starting date. If there is a company or non-TUM research institution involved, we also need the name of your external contact person.

B. For the registration: If you are supervised by a graduate student or post-doc, this person is the "Betreuer/-in / Principal advisor". Prof. Pfeffer is the "1.Prüfer/-in / First examiner".

C. External Advising: Some students want to do their thesis in collaboration with a Professor at a different university (e.g., related to a stay abroad). That is fine. This person then becomes your Principle advisor. The principal adviser will supervise the thesis and also will give a suggestion for the final grade. Per TUM-rules, Prof. Pfeffer will be the first examiner. If your principal advisor is a **TUM-external person** (or even a scholar from a TUM School other than SOT or CIT), we can only start the registration process after Prof. Pfeffer has received an email from your principal advisor stating that s/he agrees with your plans and will supervise the thesis.

3. Important Administrative Information

1. From the date of turning in the registration form (HfP) / registered starting date (CIT), you have exactly 3 months to finish your BA thesis or 6 months to finish your MA thesis. Some students/programs deviate from these timelines - only you know.
2. A BA-Thesis is 40 pages or more altogether, an MA Thesis should be 60 or more pages, including references and everything. This might sound a lot, but don't worry, with all these front pages and reference pages and figures and tables, the actual text is normally about 10 pages shorter. Feel free to add a lot of figures to your work, but be aware that this should increase the number of pages.
3. TUM has templates for the cover and the actual document:
https://portal.mytum.de/corporatedesign/index_html/vorlagen/index_wissenschaftliche_arbeiten - Download these templates now and do everything you do right away in the template. The best would be to even write your exposée in the template (without all these front pages of course).
4. You can write your thesis in Word or in Latex. Latex is nicer, but both are fine. If you use Latex, you can do it locally or, in case you have an internet connection most of the time, you can use overleaf.com or sharelatex.tum.de
5. You have to turn in (Abgabe) your final thesis at the student office or via the online portal. Sending a copy to the thesis advisor cannot replace the official submission at the student office.
6. You cannot make any changes after submission - the submitted version is the final version that will get published.
7. Extension to your deadline can only be granted in rare circumstances and must be discussed before the deadline directly with the student office.
8. The date next to the professor's signature is the starting date of the 3/6 months time period. So, don't get the signature before you are ready to start!
9. Please be aware that different departments have different rules, e.g. for when you can officially submit the thesis registration form, etc. double-check on the department's or school's website.
10. **One more time: Every formal step, i.e. registration and submission of the thesis, or request for extension has to be done with the student office or the online systems provided by your school.**

4. References

- You really need to collect and store your references in a proper and useful way from the beginning - including the Exposé.
- You need a system that you can later use to handle your references in the thesis.
- We use Zotero. If you have experience with Endnote, feel free to use it. But if you have no clue about these tools, use Zotero.
- If you want to use Zotero, install it, install the browser extension (which will make it very easy to collect the references), install the Word plugin (which will make it very easy to handle your references in the thesis). If you use Latex, you can easily export the references to bib format. Find somebody to show you the Zotero workflow.
- For the actual citations, use APA style or something very similar: (https://en.wikipedia.org/wiki/APA_style). For Computer Science and engineering students, IEEE style is fine as well.

5. Final Presentation

A. For some study programs, it is necessary to give a final presentation (e.g. computer science). If this is the case for you, be active and talk to us right after submitting your thesis, so that we can find a date for your presentation. Formally, the presentation can be at any time after you have submitted the thesis. We have our team meetings Wednesday, 9-10am, please inform your advisor which Wednesdays would be good for you. It makes sense to have the presentation 1-3 weeks after your submission.

B. When you prepare your presentation, please follow these guidelines:

- Please prepare a presentation that takes about 15 minutes.
- Focus on stuff that YOU did. There is no need for introducing the topic and literature. One slide for motivation/research questions and one slide for important literature is good enough. The rest we can find in your thesis.
- There is a good chance that you have spent a lot of time with data collection and/or data preparation. You might even have written a lot of code. That is fine. We can read about it in your thesis. We don't want to hear about all of this in the presentation. One slide for data collection, one for pre-processing in case it is important. At the same time, we need to understand your data and possible issues. If you have written code, have an overview slide about your software architecture or workflow.
- Show some slides with results. And with some, I mean 3-5 and not 20. Also, spend more time with these slides, e.g., add graphs. Make sure that you show us your main (and maybe cool) results. Again, we know that you have created 297 charts and tables, we can all see them in your thesis.
- Make sure you have a slide or two of discussion at the end. Could you answer your questions? Was the data useful? Were the methods appropriate? What would you do differently if you had to start all over again?
- And one more thing that is really important: STAY AT 15 MINUTES!